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Introduction

This report, which is a component of MATH1312 Assignment 3, uses logistic regression and multiple linear regression techniques to analyse two distinct datasets.

Using the asphalt dataset, Question 1 examines pavement durability. Using six predictors linked to asphalt composition and experimental conditions—viscosity, surface asphalt percentage, base asphalt %, fines, voids, and a run indicator—the goal is to forecast the change in rut depth (Y). A complete regression model is fitted, individual predictor significance is interpreted using t-tests, the overall model fit is assessed using ANOVA, multicollinearity is checked using VIF, and the most parsimonious model is determined by utilizing regression of all feasible subsets. Influence diagnostics and residual plots are used to verify the model's appropriateness.

Using the byssinosis dataset, Question 2 investigates the prevalence of byssinosis, a respiratory illness, among workers in the cotton sector. All five predictors—dustiness level, race, sex, smoking status, and duration of employment—are categorical, while the response variable is binary (disease present or absent). The importance of the predictors is evaluated after fitting a logistic regression model. The Hosmer-Lemeshow goodness-of-fit test, ROC curve analysis, and residual diagnostics are used to assess the model's adequacy. Lastly, disease probabilities for specific worker profiles are manually calculated using the model.

To evaluate the validity and interpretability of both models, the study uses a variety of statistical methods in R, such as regression modelling, hypothesis testing, goodness-of-fit metrics, and diagnostic assessments.

Methodology

The statistical software R was used to do the analysis, which involved applying logistic regression and multiple linear regression methods to the datasets that were supplied for the assignment.

Question 1 – Asphalt Data Analysis

The first dataset focused on predicting the **change in rut depth (Y)** based on six predictors:

- X_1: Viscosity of rut depth
- X_2: % of asphalt in the surface course
- X_3: % of asphalt in the base course
- X_4: % of fines in the surface course
- X_5: % of voids in the surface course
- X_6: Run indicator

All six predictors were first used to create a complete multiple linear regression model. The following criteria were used to evaluate the model:

- t-tests to assess each predictor's significance separately
- To evaluate the overall model significance, use ANOVA (F-test).
- Adjusted and R-squared Values of R-squared to assess model fit

The Variance Inflation Factors (VIF) were computed to account for possible multicollinearity. To find the most economical model, All Possible Subsets Regression was used to choose models based on R2 values. Among the diagnostic tests were:

- Plotting residuals against fitted data to evaluate linearity
- To assess the residuals' normality, use a Q-Q plot.
- Analyse homoscedasticity and influence using scale-location and residuals vs. leverage graphs.
- Using Cook's Distance to pinpoint significant observations

Question 2 – Byssinosis Data Analysis

The prevalence of byssinosis, a lung condition, among cotton sector workers was examined in the second dataset. The predictors were all categorical, while the response variable was binary (disease presence or absence):

- X_1: Dustiness of workplace (1 = high, 2 = medium, 3 = low)
- X_2: Race (1 = European, 2 = Other)
- X_3: Sex (1 = Male, 2 = Female)
- X_4: Smoking history (1 = Smoker, 2 = Non-smoker)
- X_5: Length of employment (1 = <10 yrs, 2 = 10–20 yrs, 3 = >20 yrs)

The glm() method is used to fit a logistic regression model with a binomial family. The model was interpreted using:

- P-values and estimated coefficients for every predictor
- Exponentiated coefficients to interpret odds ratios

Model adequacy was assessed using:

- Hosmer-Lemeshow test to evaluate goodness of fit
- ROC curve and AUC to measure classification performance
- Deviance residual plots to identify potential model issues

Finally, the fitted model was utilized to manually compute probabilities for specific combinations of predictor levels using the logistic function. All statistical findings were interpreted in context, and the results were backed by graphical and numerical data.

Result

Question 1

Part A

The full multiple linear regression model was fitted using six predictors:

$$y=\beta0+\beta1x1+\beta2x2+\beta3x3+\beta4x4+\beta5x5+\beta6x6+\varepsilon$$

Here

x_1 = Viscosity of rut depth

x_2 = % of asphalt in the surface course

x_3 = % of asphalt in the base course

x_4 = % of fines in the surface course

x_5 = % of voids in the surface course

x_6 = Run indicator

y = Change in rut depth

```
Call:
lm(formula = y ~ x1 + x2 + x3 + x4 + x5 + x6, data = asphalt)

Residuals:
    Min      1Q  Median      3Q     Max
-5.6781 -1.8309  0.1751  1.4858 11.1262

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -57.584153  35.896498  -1.604   0.1218
x1            0.003071   0.008161   0.376   0.7100
x2            7.498028   3.967155   1.890   0.0709 .
x3            6.225817   4.812723   1.294   0.2081
x4            0.522211   1.174673   0.445   0.6606
x5           -0.241275   1.684963  -0.143   0.8873
x61         -10.772593   1.970769  -5.466 1.28e-05 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.94 on 24 degrees of freedom
Multiple R-squared:  0.7274,    Adjusted R-squared:  0.6592
F-statistic: 10.67 on 6 and 24 DF,  p-value: 8.588e-06
```

Figure 1: Model Summary

From the output we get

Intercept = -57.5841

x_1 (Viscosity) = 0.0030

x_2 (Surface asphalt %) = 7.4980

x_3 (Base asphalt %) = 6.2258

x_4 (Fines %) = 0.5222

x_5 (Voids %) = -0.2412

x_6 (Run indicator) = -10.7725

The fitted regression equation comes out to be:

$$\hat{y} = -57.5841 + 0.0031x1 + 7.4980x2 + 6.2258x3 + 0.5222x4 - 0.2413x5 - 10.7725x6 + \varepsilon$$

Inference: The multiple linear regression model for predicting rut depth change (y) using six predictors—viscosity, surface asphalt (%), base asphalt (%), fines (%), voids (%), and run indicator—is statistically significant, with an F-statistic of 10.67 and a p-value of 8.588×10^{-6}. The model accounts for 72.7% of the variation in rut depth ($R^2 = 0.7274$), with an adjusted R^2 of 0.6592, indicating a strong fit. The run indicator (x_6) is the only highly significant predictor ($p < 0.001$), suggesting a strong influence on the response. The surface asphalt percentage (x_2) is marginally significant ($p = 0.0709$), hinting at a possible effect. The remaining predictors—viscosity, base asphalt, fines, and voids—are not statistically significant ($p > 0.2$), indicating limited individual contributions. These results suggest that the model performs well overall, but simplification may be possible by removing non-significant variables in further refinement.

```
> anova(full_model_asphalt)
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value    Pr(>F)    
x1         1 424.93  424.93 27.3767 2.310e-05 ***
x2         1  16.13   16.13  1.0393    0.3182    
x3         1  28.40   28.40  1.8295    0.1888    
x4         1  19.73   19.73  1.2711    0.2707    
x5         1  41.05   41.05  2.6450    0.1169    
x6         1 463.77  463.77 29.8792 1.283e-05 ***
Residuals 24 372.51   15.52                      
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 2: ANOVA Table of the Model.

ANOVA hypothesis:

H_0, $\beta i = 0$: The predictor has no effect on the response variable (its regression coefficient is equal to zero).

H_1, $\beta i \mathrel{!}= 0$: The predictor has a significant effect on the response variable (its regression coefficient is not equal to zero).

Inference: The ANOVA table shows that x_1 (viscosity) and x_6 (run indication) are highly significant, with p-values of 2.31×10^{-5} and 1.28×10^{-5}, respectively ($p < 0.001$). The small p-values strongly support rejecting the null hypothesis for x_1 and x_6, demonstrating a statistically significant effect on rut depth. The remaining predictors — x_2 (surface asphalt), x_3 (base asphalt), x_4 (fines), and x_5 (voids) — have p-values greater than 0.1, indicating that we fail to reject the null hypothesis for these variables. Their contributions to explaining response variation are not statistically significant when other variables are present. The model's most significant predictors are the viscosity of rut depth (x_1) and the run condition (x_6).

Part B

```
Fitted equation when run = 1:
> cat(paste0("y = ",
+            round(intercept_run1, 4), " + ",
+            round(coefs["x1"], 4), "*x1 + ",
+            round(coefs["x2"], 4), "*x2 + ",
+            round(coefs["x3"], 4), "*x3 + ",
+            round(coefs["x4"], 4), "*x4 + ",
+            round(coefs["x5"], 4), "*x5\n\n"))
y = -68.3567 + 0.0031*x1 + 7.498*x2 + 6.2258*x3 + 0.5222*x4 + -0.2413*x5
```

Figure 3: Model with run=1.

```
Fitted equation when run = -1:
> cat(paste0("y = ",
+            round(intercept_runm1, 4), " + ",
+            round(coefs["x1"], 4), "*x1 + ",
+            round(coefs["x2"], 4), "*x2 + ",
+            round(coefs["x3"], 4), "*x3 + ",
+            round(coefs["x4"], 4), "*x4 + ",
+            round(coefs["x5"], 4), "*x5\n"))
y = -46.8116 + 0.0031*x1 + 7.498*x2 + 6.2258*x3 + 0.5222*x4 + -0.2413*x5
```

Figure 4: Model with run= -1.

Fitted equation for y

When x_6 (run) = 1:

$$\hat{y} = -68.3567 + 0.0031x_1 + 7.4980x_2 + 6.2258x_3 + 0.5222x_4 - 0.2413x_5 + \varepsilon$$

When x_6 (run) = −1:

$$\hat{y} = -46.8116 + 0.0031x_1 + 7.4980x_2 + 6.2258x_3 + 0.5222x_4 - 0.2413x_5 + \varepsilon$$

Inference: The slope coefficients are the same in both fitted equations, suggesting that the association between each predictor and rut depth remains constant regardless of run situation. The intercept differs significantly between the two models: −68.3567 when run = 1 and −46.8116 when run = −1. The 21.55-unit change in intercepts indicates a large baseline variation in rut depth depending on the experimental run. The run indicator (x_6) significantly explains changes in rut depth, highlighting its importance in the regression model.

Part C

```
> summary_r
Subset selection object
Call: regsubsets.formula(y ~ x1 + x2 + x3 + x4 + x5 + x6, data = asphalt)
6 Variables  (and intercept)
    Forced in Forced out
x1      FALSE      FALSE
x2      FALSE      FALSE
x3      FALSE      FALSE
x4      FALSE      FALSE
x5      FALSE      FALSE
x61     FALSE      FALSE
1 subsets of each size up to 6
Selection Algorithm: exhaustive
         x1  x2  x3  x4  x5  x61
1  ( 1 ) " " " " " " " " " " "*"
2  ( 1 ) " " "*" " " " " " " "*"
3  ( 1 ) " " "*" "*" " " " " "*"
4  ( 1 ) " " "*" "*" "*" " " "*"
5  ( 1 ) "*" "*" "*" "*" " " "*"
6  ( 1 ) "*" "*" "*" "*" "*" "*"
> # View R² values for reference
> print(summary_r$rsq)
[1] 0.6575583 0.6946994 0.7239215 0.7257687 0.7271661 0.7273990
> # Identify best model based on maximum R²
> best_rsq_index <- which.max(summary_r$rsq)
> best_rsq_index
[1] 6
```

Figure 5: Summary of Subset Selection and R square values of each model.

The subset selection method was used to determine the best regression model for predicting rut depth (y) by combining six predictors (x1–x6). The selection criterion in the question was the greatest R^2, which measures the proportion of variance explained by the model.

Inference:

The maximum R^2 value is 0.7274, which corresponds to the model including all six predictors:

- x1 (viscosity), x2 (surface asphalt%), x3 (base asphalt%), x4 (fines%), x5 (voids%), and x6 (run indicator).
- This model was discovered at a subset size of 6 (the entire model).
- Adding predictors enhances explanatory power, as evidenced by the increasing improvement in R^2 with subset size. However, the benefit is minor beyond 4-5 variables.

Best Model based on R^2

```
> summary(best_model)

Call:
lm(formula = reduced_formula, data = asphalt)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.6781 -1.8309  0.1751  1.4858 11.1262 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -57.584153  35.896498  -1.604   0.1218    
x1            0.003071   0.008161   0.376   0.7100    
x2            7.498028   3.967155   1.890   0.0709 .  
x3            6.225817   4.812723   1.294   0.2081    
x4            0.522211   1.174673   0.445   0.6606    
x5           -0.241275   1.684963  -0.143   0.8873    
x61         -10.772593   1.970769  -5.466 1.28e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.94 on 24 degrees of freedom
Multiple R-squared:  0.7274,	Adjusted R-squared:  0.6592 
F-statistic: 10.67 on 6 and 24 DF,  p-value: 8.588e-06
```

Figure 6: Summary of the best model based on R^2.

Inference: The model with the highest R^2 includes all six predictors: x1 (viscosity), x2 (surface asphalt%), x3 (base asphalt%), x4 (fines%), x5 (voids%), and x6 (run indicator).

The fitted regression equation is:

$$\mathbf{yhat=-57.5841+0.0031x1+7.4980x2+6.2258x3+0.5222x4-0.2413x5-10.7725x6+\varepsilon}$$

The model with the highest R^2 score incorporates all six factors and performs well overall. The model accounts for 72.74% of rut depth variability ($R^2 = 0.7274$) and is statistically significant (F-statistic = 10.67, p-value = 8.588×10^{-6}). The run indicator is the only highly significant predictor ($p < 0.001$), indicating a considerable impact on rut depth from the experimental run. The surface asphalt %

is marginally significant (p = 0.0709), indicating a possible effect, but the other variables—viscosity, base asphalt, fines, and voids—are not statistically significant (p > 0.2). Despite including all predictors, the model maintains significant explanatory power and fits with the selection criterion based on R^2. The selected model is equal to the original full model, demonstrating that no simpler model gave a better R^2. Therefore, the full model remains the best choice under this criterion.

Diagnostic Tests

Residual Analysis

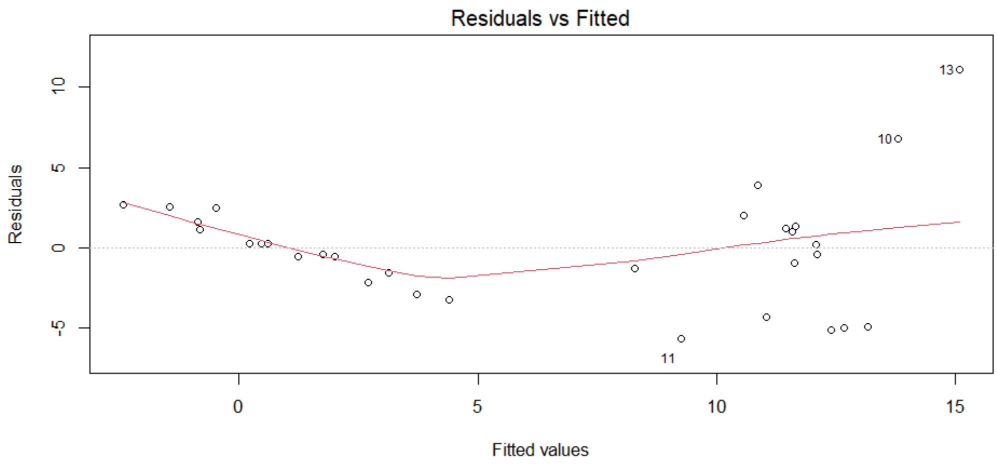

Figure 7:Residuals vs Fitted Plot.

Inference: The Residuals vs Fitted plot evaluates linearity and homoscedasticity. The residuals show a slight curved trend, indicating modest departures from the completely linear connection. There is also a slight increase in the spread of residuals as the fitted values grow, indicating minor heteroscedasticity. Notably, observations 10, 11, and 13 appear to differ significantly from the rest, implying possible outliers. Despite these minor problems, the plot shows no serious violations, indicating that the model assumptions are reasonable but might be improved.

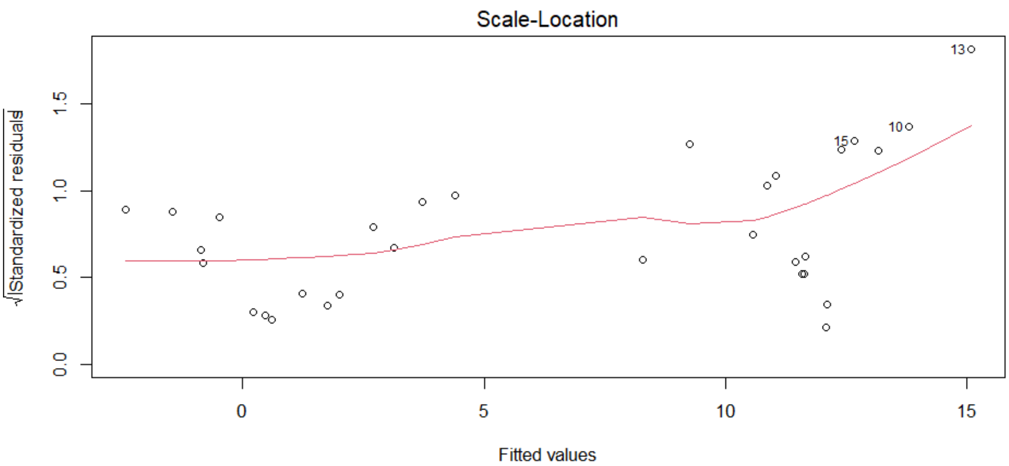

Figure 8:Scale-Location plot.

Inference: The Scale-Location plot examines the assumption of homoscedasticity (constant variance). In the plot, the red trend line has a moderate upward slope, showing a slight increase in residual spread as fitted values increase. This shows the presence of minor heteroscedasticity, though not severe. Observations 10, 13, and 15 continue to stand out with higher standardized residuals and may require more investigation. The variance is not constant, but it is moderate, and the assumption of homoscedasticity is only slightly broken.

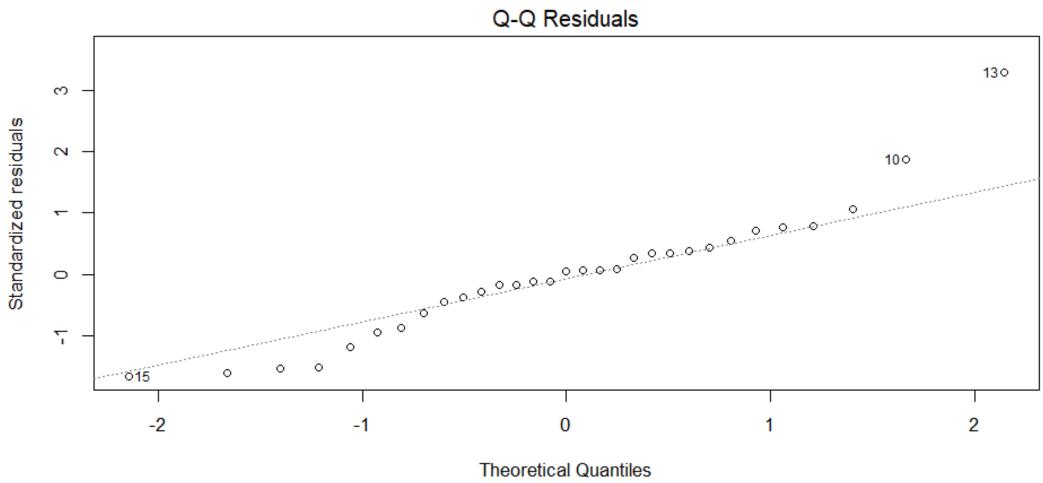

Figure 9: Q-Q Plot.

Inference: The Q-Q plot shows that most residuals lie close to the reference line, implying that the residuals are roughly normally distributed. There are some variations at the extremes, especially for observations 10, 13, and 15, which appear to be moderate outliers. However, these deviations are not significant, and the overall pattern confirms the normality assumption. Thus, the model residuals show an adequate level of normality.

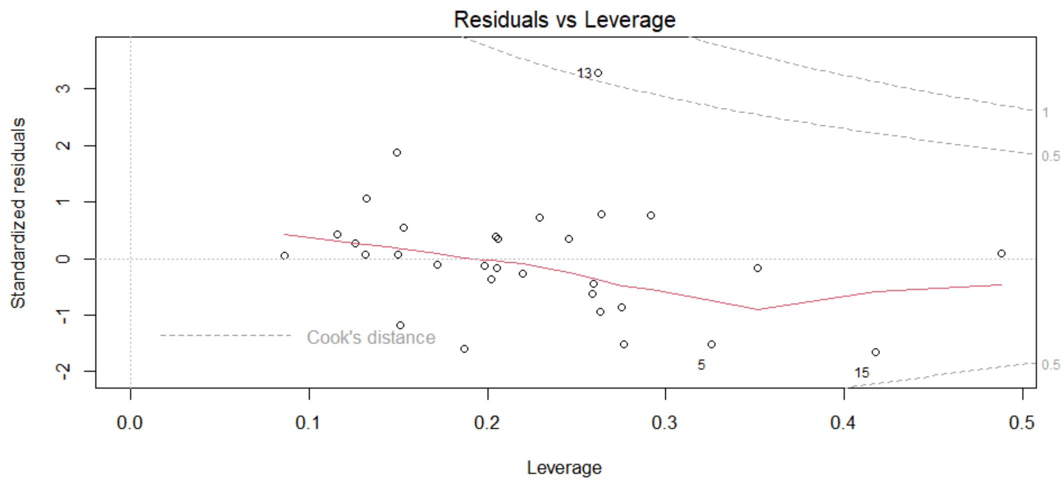

Figure 10: Residual vs Leverage Plot.

Inference: The Residuals vs Leverage graph is used to identify data points that have a significant impact on the regression model. In this map, most data points are well within the Cook's distance contours, showing little influence. Observation 13 exhibits moderately high leverage and standardized residuals, putting it toward the outer edge of Cook's distance lines, indicating some effect. Observations 5 and 15 have slightly more leverage, but their influence is limited. Overall, there is not much proof that any observation has a significant impact on the model, indicating its stability.

Normality Test

H_0: The residuals are normally distributed.

H_1: The residuals are not normally distributed.

```
> shapiro.test(residuals(best_model))

        Shapiro-Wilk normality test

data:  residuals(best_model)
W = 0.93011, p-value = 0.04416

> # Anderson-Darling
> ad.test(residuals(best_model))

        Anderson-Darling normality test

data:  residuals(best_model)
A = 0.56804, p-value = 0.1292
```

Figure 11: Shapiro-Wilk and Anderson-Darling Test.

Inference: The Shapiro-Wilk test resulted in a W statistic of 0.9301 and a p-value of 0.0442, which is less than the 0.05 significance level. As a result, we reject the null hypothesis, indicating that the residuals deviate slightly from normality. However, the Anderson-Darling test returned a p-value of 0.1292, which is greater than 0.05. As a result, we fail to reject the null hypothesis, indicating no substantial deviation from normality.
The evidence for both tests is mixed when taken together. The Shapiro-Wilk test indicates probable mild non-normality, whereas the Anderson-Darling test does not. Overall, the residuals appear to be quite normal, with only small deviations that do not provide significant evidence against the normality assumption.

Homoscedasticity Test

H_0: The residuals have constant variance (homoscedasticity).

H_1: The residuals have non-constant variance (heteroscedasticity)

```
> # Homoscedasticity Test
> ncvTest(best_model)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 12.3959, Df = 1, p = 0.00043028
```

Figure 12: NCV Test.

Inference: The Non-constant Variance Score Test (ncvTest) produced a Chi-square statistic of 12.3959, 1 degree of freedom, and p-value of 0.00043. Since the p-value is significantly lower than 0.05, we reject the null hypothesis. This is significant evidence of heteroscedasticity, which means that the variance of residuals is not constant.
This breach of the homoscedasticity assumption may have an impact on the reliability of standard errors and inference, indicating that the model could benefit from corrective methods such as robust standard errors or response variable transformation.

Autocorrelation

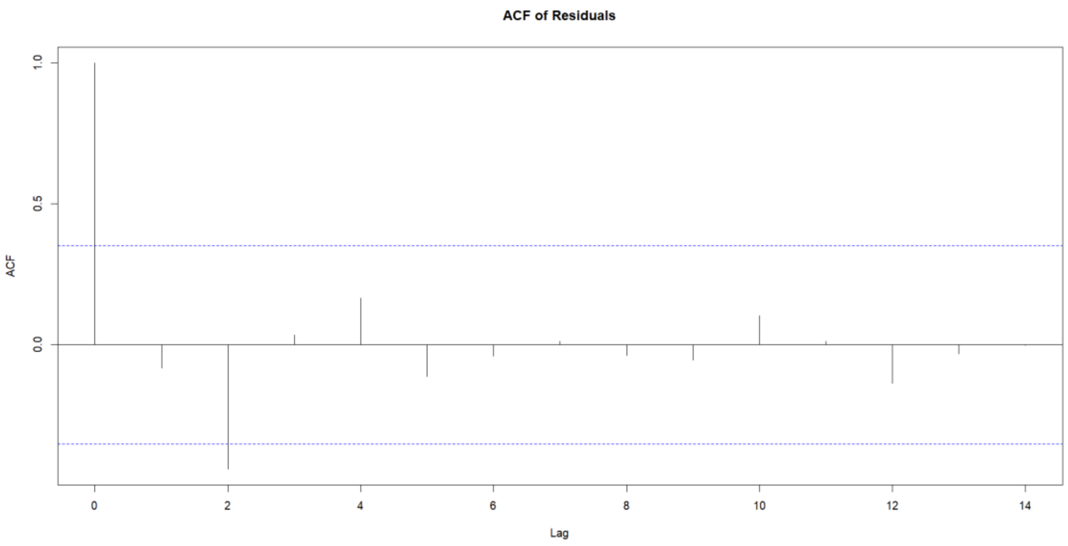

Figure 13: ACF Plot of Residuals

Inference: The ACF (Autocorrelation Function) plot of residuals indicates that most autocorrelation values fall within the blue significance boundaries, with no bars surpassing the thresholds except for lag 0. This indicates that there is no significant autocorrelation in the residuals.
The lack of pattern or huge spikes in the lags indicates that the residuals are independent of one another, which supports the regression model's assumption of residual independence.

```
> dwtest(best_model)

        Durbin-Watson test

data:  best_model
DW = 2.0982, p-value = 0.4996
alternative hypothesis: true autocorrelation is greater than 0
```

Figure 14: Durbin-Watson Test.

H_0: There is no autocorrelation in the residuals.

H_1: There is positive autocorrelation in the residuals.

Inference: The Durbin-Watson statistic is 2.0982, with a corresponding p-value of 0.4996, which exceeds the 0.05 level of significance. Therefore, we fail to reject the null hypothesis. This shows that the residuals include no significant evidence of positive autocorrelation, and the regression model's independence assumption is reasonably satisfied.

Multicollinearity Test

```
> # VIF
> vif(best_model)
      x1       x2       x3       x4       x5       x6 
2.039971 1.441857 1.393992 1.417067 1.832991 1.937266 
```

Figure 15: VIF Score.

Inference: The Variance Inflation Factor (VIF) values for all predictors in the model range from 1.39 to 2.04, which is significantly lower than the typically accepted threshold of 10. This suggests that multicollinearity is not an issue in the fitted regression model. The predictors are not substantially correlated, implying that the coefficient estimates are consistent and accurate. As a result, the model's interpretability and the validity of hypothesis testing are preserved.

Conclusion:

The model with the greatest R^2 value (0.7274) has substantial explanatory power and statistical significance, indicating a good match to the data. However, diagnostic checks indicate various issues that have an impact on the model's appropriateness. Residual plots exhibit moderate curvature and increasing variation, indicating mild nonlinearity and heteroscedasticity. This is supported by the NCV test, which finds significant statistical evidence ($p = 0.00043$) of non-constant variance, contradicting the homoscedasticity assumption.

The Anderson-Darling test ($p = 0.1292$) shows no significant deviation from normality in residuals, although the Shapiro-Wilk test ($p = 0.0442$) detects a minor departure. Importantly, autocorrelation is not an issue, as evidenced by the ACF plot and Durbin-Watson test ($p = 0.4996$). Multicollinearity is also not an issue, as all VIF values are much below the acceptable threshold of 10.

In conclusion, while the model is statistically robust and contains no severe violations of essential assumptions, the presence of heteroscedasticity shows that improvements are possible. Applying robust standard errors or altering the response variable may improve the reliability of inferences. Nonetheless, the model remains reasonably effective at predicting rut depth.

Question 2

Part A

```
> summary(model_logit)

Call:
glm(formula = cbind(BysYes, BysNo) ~ Dust + Race + Sex + Smoke +
    Employ, family = binomial, data = byssinosis)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.9452     0.2334  -8.334  < 2e-16 ***
Dust2        -2.5799     0.2921  -8.834  < 2e-16 ***
Dust3        -2.7306     0.2153 -12.681  < 2e-16 ***
Race2         0.1163     0.2072   0.562 0.574426
Sex2          0.1239     0.2288   0.542 0.587983
Smoke2       -0.6413     0.1944  -3.299 0.000971 ***
Employ2       0.5641     0.2617   2.156 0.031091 *
Employ3       0.7531     0.2161   3.484 0.000493 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 322.527  on 64  degrees of freedom
Residual deviance:  43.271  on 57  degrees of freedom
AIC: 165.95

Number of Fisher Scoring iterations: 5
```

Figure 16: Summary of logistic regression model.

A logistic regression model was used to determine the impact of workplace and demographic characteristics on the chance of workers contracting byssinosis, a respiratory ailment common in cotton industry workers. The response variable was modelled using the event/trial format (cbind(BysYes, BysNo)), with five categorical predictors: Dust, Race, Sex, Smoke, and Employ.

Inference:

From the R output we find:

Intercept = −1.9452

Dust2 = −2.5799

Dust3 = −2.7306

Race2 = 0.1163

Sex2 = 0.1239

Smoke2 = −0.6413

Employ2 = 0.5641

Employ3 = 0.7531

The fitted logistic regression equation is:

$$\log(p/(1-p)) = -1.9452 - 2.5799 * Dust2 - 2.7306 * Dust3 + 0.1163 * Race2 + 0.1239 * Sex2 - 0.6413 * Smoke2 + 0.5641 * Employ2 + 0.7531 * Employ3 + \varepsilon$$

Here:

p is the probability that a worker suffers from byssinosis.

Dust exposure significantly predicts byssinosis, with both medium (Dust2) and low (Dust3) exposure levels having extremely low p-values ($p < 2e\text{-}16$), indicating a substantial reduction in disease risk compared to high exposure. Smoking status (Smoke2) is also important ($p = 0.000971$), with nonsmokers having a lower risk of getting the disease. Employment duration has a major influence as well—those employed for 10-20 years (Employ2) and more than 20 years (Employ3) had significantly elevated risk ($p = 0.0311$ and $p = 0.000493$, respectively), indicating that cumulative exposure increases vulnerability. In contrast, race and sex are not statistically significant, with p-values of 0.574 and 0.588, showing minimal independent influence on disease likelihood. The model has an excellent fit, as evidenced by a significant drop in deviation from 322.527 to 43.271 and an AIC of 165.95. Overall, the model successfully identifies dust levels, smoking habits, and employment duration as critical risk factors for byssinosis, emphasizing the importance of occupational and lifestyle factors on worker health.

```
> # Exponentiated coefficients (odds ratios)
> exp(coef(model_logit))
(Intercept)       Dust2       Dust3       Race2        Sex2      Smoke2     Employ2     Employ3 
 0.14295886  0.07578099  0.06517821  1.12338740  1.13193648  0.52658281  1.75788376  2.12365346 
```

Figure 17: Exponentiated Coefficients.

Inference: The logistic regression model's exponentiated coefficients show the odds ratios for each predictor level compared to its baseline. The odds ratios for dust exposure are 0.0758 for medium (Dust2) and 0.0655 for low (Dust3), showing that the chances of getting byssinosis are around 92%-93% lower than workers exposed to high dust levels. Smoking status (Smoke2) had an odds ratio of 0.5266, indicating that nonsmokers have a 47% lower risk of acquiring byssinosis than smokers. individuals with 10-20 years of experience in the industry (Employ2) had 1.76 times the odds of disease, while those with more than 20 years (Employ3) had 2.12 times the risk, compared to individuals with fewer than 10 years. In contrast, Race2 (odds ratio = 1.123) and Sex2 (1.132) have minimal impacts, with odds ratios near to one, consistent with previous findings of statistical insignificance. Overall, these odds ratios support the hypothesis that dust exposure, smoking, and long-term employment have a significant impact on the risk of byssinosis.

```
> exp(confint(model_logit))
Waiting for profiling to be done ...
                 2.5 %     97.5 %
(Intercept) 0.08942893 0.22340848
Dust2       0.04159539 0.13129952
Dust3       0.04230407 0.09854175
Race2       0.74834967 1.68708592
Sex2        0.71847253 1.76552092
Smoke2      0.35601536 0.76430677
Employ2     1.03958048 2.90981429
Employ3     1.39457713 3.25709224
```

Figure 18: Confidence Interval

Inference:

Dust Exposure:
The odds ratio (CI) for Dust2 (medium exposure) is (0.0416, 0.1313), whereas Dust3 (low exposure) is (0.0423, 0.0985). Since both ranges are totally below one, this demonstrates that medium and low dust exposures considerably reduce the odds of byssinosis when compared to high exposure.

Smoking (Smoke2):
The CI (0.3561, 0.7643) is completely below one, indicating that nonsmokers are much less likely to develop the condition than smokers.

Employment Duration:
Employ2 has a CI of (1.0396, 2.9098), whereas Employ3 has a CI of (1.3946, 3.2571), both of which are more than 1. This suggests that longer employment relates to a much-increased risk of acquiring byssinosis.

Race and Sex:
Race2 (CI: 0.7483, 1.6871) and Sex2 (CI: 0.7185, 1.7655) both have intervals of one, indicating that there is no statistically significant effect on illness odds.

Overall, the confidence intervals support earlier findings: dust exposure, smoking, and employment duration are significant predictors, whereas race and sex are not.

Part B

<u>Chi-square goodness-of-fit test</u>

```
> pchisq(deviance(model_logit), df.residual(model_logit), lower.tail = FALSE)
[1] 0.9103186
```

Figure 19: Chi-Square Goodness of Fit Test Value.

H_0: The logistic regression model fits the data adequately.

H_1: The model does not fit the data adequately.

Inference: The p-value for the goodness-of-fit test using the deviance statistic is 0.9103, which is significantly greater than 0.05. This suggests that we fail to reject the null hypothesis, which states that the model fits the data well. In other words, the observed data are not significantly different from the model's predictions. Thus, the logistic regression model has a good overall fit, showing that the model's structure and predictors are acceptable for describing the outcome variable.

<u>ROC Curve</u>

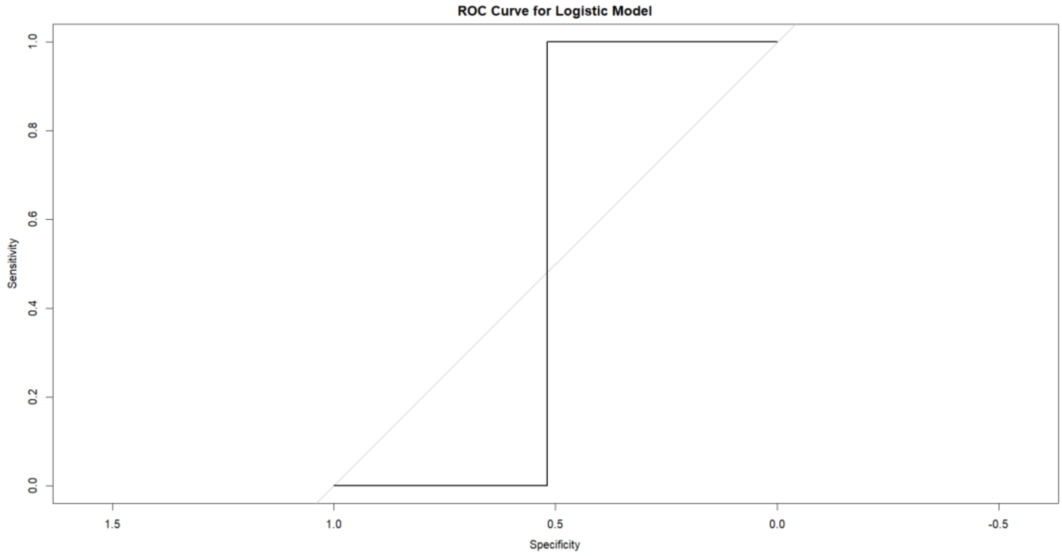

Figure 20: ROC Curve.

```
> auc(roc_obj)
Area under the curve: 0.5185
```

Figure 21: Area Under the Curve Value.

Inference: The ROC (Receiver Operating Characteristic) curve shows how the logistic regression model trades off sensitivity and specificity. The area under the curve (AUC) is 0.5185, only slightly better than random guessing (AUC = 0.5). This low AUC indicates that, while the model is statistically significant and fits the data well, its discriminative capacity to correctly categorize persons with and without byssinosis is quite limited. As a result, while the model describes the relationship between predictors and outcomes, it performs poorly in predicting actual class labels, indicating that further modification or additional variables are required to enhance classification performance.

Deviance Residual Plot

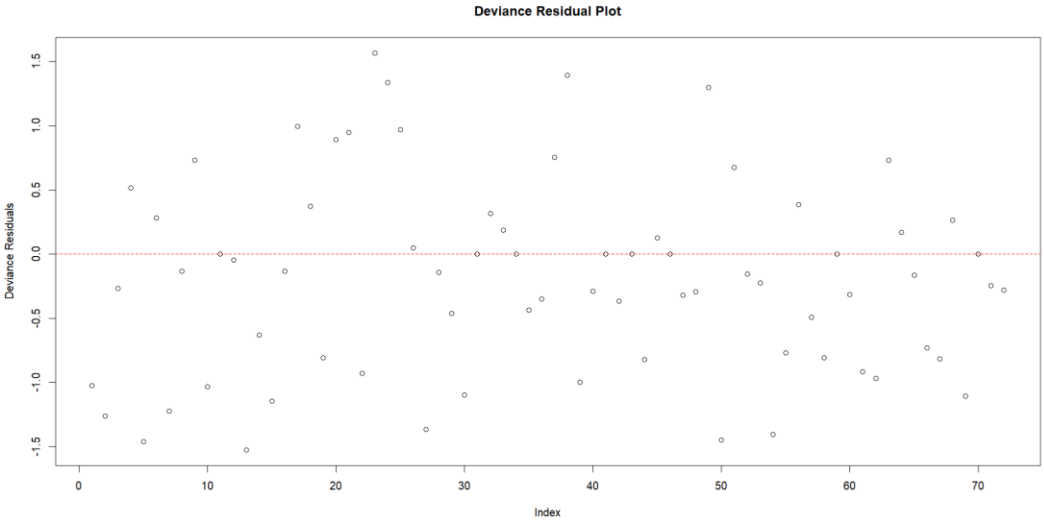

Figure 22: Deviance Residual Plot.

Inference: The Deviance Residual Plot shows residuals distributed about the horizontal reference line at zero, without any discernible pattern or structure. This shows that the model fits the data reasonably well and that the logistic regression model's assumptions have not been violated. There is no evidence of systematic bias or model misspecification. The residuals appear symmetrically distributed with a consistent spread across observations, indicating that the model is adequate for capturing the relationship between predictors and response variables.

Conclusion from the diagnostic tests:

Several diagnostic evaluations show that the logistic regression model has excellent statistical adequacy. The Chi-square goodness-of-fit test yielded a high p-value (0.9103), showing that there is no significant difference between the observed and projected values, implying that the model fits the data well. The deviance residual plot displays no obvious trends or outliers, which supports the model's accuracy and the absence of systematic bias.

However, while the model fits the data structurally, its predictive power is restricted. The ROC curve yielded an AUC of 0.5185, which is scarcely better than random guessing. This demonstrates that, while the model captures relationships between predictors and the result (byssinosis), it lacks the power to correctly categorize individuals.

Finally, while the model adequately explains the relationships within the data, it falls short in terms of classification accuracy, underlining the need for model refinement or additional predictive factors to improve its practical utility.

Part C

The fitted logistic regression equation is:

$$\log(p/(1-p)) = -1.9452 - 2.5799 * Dust2 - 2.7306 * Dust3 + 0.1163 * Race2 + 0.1239 * Sex2 - 0.6413 * Smoke2 + 0.5641 * Employ2 + 0.7531 * Employ3 + \varepsilon$$

To solve the equation

x1=Dust

x2=Race

x3=Sex

x4=Smoke

x5=Employ

Probability will be calculated using:

$$p=e^{logit}/(1+e^{logit})$$

Reference levels (baseline) = 0 for dummy variables

Solve for xl = 2, x2 = 2, x3 = 1, x4 = 2, x5 = 3

Dust2 = 1, Dust3 = 0

Race2 = 1

Sex2 = 0

Smoke2 = 1

Employ2 = 0, Employ3 = 1

$$\begin{aligned} logit &= -1.9452-2.5799(1)+0+0.1163(1)+0+(-0.6413)(1)+0+0.7531(1) \\ &= -1.9452-2.5799+0.1163-0.6413+0.7531 \\ &= -4.297 \end{aligned}$$

$$\begin{aligned} p &= e^{(-4.297)}/(1+e^{(-4.297)}) \\ &\approx 0.0136/(1+0.0136) \\ &\approx \mathbf{0.0134} \end{aligned}$$

Solve for xl = 1, x2 = 2, x3 = 2, x4 = 1, x5 = 3

Dust2 = 0, Dust3 = 0

Race2 = 1

Sex2 = 1

Smoke2 = 0

Employ2 = 0, Employ3 = 1

$$\begin{aligned} logit &= -1.9452+0+0.1163(1)+0.1239(1)+0+0+0.7531(1) \\ &= -1.9452+0.1163+0.1239+0.7531 \\ &= -0.952 \end{aligned}$$

$$\begin{aligned} p &= e^{(-0.952)}/(1+e^{(-0.952)}) \\ &\approx 0.3857/(1+0.3857) \\ &\approx \mathbf{0.2784} \end{aligned}$$

Solve for xl = 2, x2 = 1, x3 = 1, x4 = 2, x5 = 2

Dust2 = 1, Dust3 = 0

Race2 = 0

Sex2 = 0

Smoke2 = 1

Employ2 = 1, Employ3 = 0

$\text{logit} = -1.9452 - 2.5799(1) + 0 - 0.6413(1) + 0.5641(1) + 0$

$= -1.9452 - 2.5799 - 0.6413 + 0.5641$

$= -4.6023$

$p = e^{(-4.6023)} / (1 + e^{(-4.6023)})$

$\approx 0.0101 / (1 + 0.0101)$

$\approx$ **0.00999**

Solve for xl = 3, x2 = 1, x3 = 2, x4 = 2, x5 = 1

Dust2 = 0, Dust3 = 1

Race2 = 0

Sex2 = 1

Smoke2 = 1

Employ2 = 0, Employ3 = 0

$\text{logit} = -1.9452 - 2.7306(1) + 0 + 0.1239(1) - 0.6413(1) + 0 + 0$

$= -1.9452 - 2.7306 + 0.1239 - 0.6413$

$= -5.1932$

$p = e^{(-5.1932)} / (1 + e^{(-5.1932)})$

$\approx 0.0055 / (1 + 0.0055)$

$\approx$ **0.0055**

Conclusion

The current study used two statistical modelling techniques—multiple linear regression and logistic regression—to evaluate datasets with practical applications: pavement durability and occupational health.

In Question 1, a multiple linear regression model was created to determine how different asphalt mixture parameters influence rut depth in pavements. The model, which included six predictors, showed a solid fit ($R^2 = 0.7274$). The run indicator (x_6) was the only highly significant variable ($p < 0.001$), indicating a considerable shift in rut depth across multiple experimental runs. The subset regression analysis, based on R^2 values, demonstrated that no simpler model outperformed the whole model, indicating its fitness for the specified criterion.

However, diagnostic evaluations discovered a few difficulties. The residual plots revealed minor heteroscedasticity and possibly non-linearity, as confirmed by the NCV test ($p = 0.00043$), but no autocorrelation or multicollinearity was seen. Normality tests yielded conflicting results—the Shapiro-Wilk test indicated a minor deviation, but the Anderson-Darling test did not. These diagnostics indicate that, while the model is statistically sound and explains a significant percentage of the variance in rut depth, changes such as using robust standard errors or transformations may improve inferential reliability.

In Question 2, a logistic regression model was used to investigate the factors that influence the risk of developing byssinosis among cotton sector workers. Dust exposure, smoking status, and employment length were significant predictors ($p < 0.05$), with considerable effects in odds ratios and confidence intervals. In contrast, race and sex were not statistically significant, indicating that they had little influence in the setting of this dataset.

The model showed good structural fit, as evidenced by a significant decrease in deviance and a high p-value (0.9103) in the Chi-square goodness-of-fit test. However, its predictive performance was insufficient, with an AUC of 0.5185, which was just slightly better than random chance. This means that, while the model captures relevant correlations, it is less effective at accurately identifying individual cases, indicating the need for feature engineering or new predictors.

Overall, both models offered useful insights in their situations. The multiple linear regression model successfully identified critical variables influencing rut depth, however it needs to be refined further to meet all assumptions. The logistic regression model revealed important risk variables for byssinosis with high reliability, but its predictive accuracy might be improved. These analyses demonstrate the efficacy of regression techniques in understanding complicated real-world occurrences, while also emphasizing the significance of model validation and diagnostic testing in reaching solid findings.

Reference

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Appendix

```
> # Question 1
> # Load the dataset
> asphalt <- read.csv("asphalt.csv")
> head(asphalt)
      y visc surf base fines voids run
1  6.75  2.8 4.68 4.87   8.4 4.916  -1
2 13.00  1.4 5.19 4.50   6.5 4.563  -1
3 14.75  1.4 4.82 4.73   7.9 5.321  -1
4 12.60  3.3 4.85 4.76   8.3 4.865  -1
5  8.25  1.7 4.86 4.95   8.4 3.776  -1
6 10.67  2.9 5.16 4.45   7.4 4.397  -1
> tail(asphalt)
      y visc surf base fines voids run
26 0.47  500 4.83 4.60   6.7 5.471   1
27 0.33  180 4.66 4.72   7.2 4.602   1
28 0.26  270 4.67 4.50   6.3 5.043   1
29 0.76  170 4.72 4.70   6.8 5.075   1
30 0.80   98 5.00 5.07   7.2 4.334   1
31 2.00   35 4.70 4.80   7.7 5.705   1
> # Rename variables in the asphalt dataset
> colnames(asphalt) <- c("y", "x1", "x2", "x3", "x4", "x5", "x6")
> # Check the new column names
> colnames(asphalt)
[1] "y"  "x1" "x2" "x3" "x4" "x5" "x6"
> # Convert x6 (run indicator) to a factor
> asphalt$x6 <- factor(asphalt$x6)
```

```
> # Question 2
> # Load the dataset
> byssinosis <- read.csv("byssinosis.csv")
> # Convert predictors to factors (all are categorical)
> byssinosis$Dust   <- factor(byssinosis$Dust)   # 1 = high, 2 = medium, 3 = low
> byssinosis$Race   <- factor(byssinosis$Race)   # 1 = European, 2 = Other
> byssinosis$Sex    <- factor(byssinosis$Sex)    # 1 = Male, 2 = Female
> byssinosis$Smoke  <- factor(byssinosis$Smoke)  # 1 = Smoker, 2 = Non-smoker
> byssinosis$Employ <- factor(byssinosis$Employ) # 1, 2, 3 = <10, 10-20, >20 yrs
> # View structure
> str(byssinosis)
'data.frame':   72 obs. of  8 variables:
 $ BysYes: int  3 0 2 25 0 3 0 1 3 2 ...
 $ BysNo : int  37 74 258 139 88 242 5 93 180 22  ...
 $ Total : int  40 74 260 164 88 245 5 94 183 24  ...
 $ Dust  : Factor w/ 3 levels "1","2","3": 1 2 3 1 2 3 1 2 3 1  ...
 $ Race  : Factor w/ 2 levels "1","2": 1 1 1 2 2 2 1 1 1 2  ...
 $ Sex   : Factor w/ 2 levels "1","2": 1 1 1 1 1 1 2 2 2 2  ...
 $ Smoke : Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1  ...
 $ Employ: Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1  ...
```

R Codes Used

To clean the R environment

rm(list = ls())

Loading necessary libraries

library(tidyverse)

```
library(car)
library(leaps)
install.packages("ResourceSelection")
library(ResourceSelection)
install.packages("pROC")
library(pROC)
library(lmtest)
library(nortest)

# Setting up the directory
setwd("C:/Users/shamb/OneDrive/Desktop/Regression Analysis/Assignment 3")

# Question 1
# Load the dataset
asphalt <- read.csv("asphalt.csv")
head(asphalt)
tail(asphalt)
# Rename variables in the asphalt dataset
colnames(asphalt) <- c("y", "x1", "x2", "x3", "x4", "x5", "x6")
# Check the new column names
colnames(asphalt)
# Convert x6 (run indicator) to a factor
asphalt$x6 <- factor(asphalt$x6)

# Question 1 Part A
# Fit full linear model
full_model_asphalt <- lm(y ~ x1 + x2 + x3 + x4 + x5 + x6, data = asphalt)
# View model summary
summary(full_model_asphalt)
# ANOVA table for overall fit
anova(full_model_asphalt)

# Question 1 Part B
# Get model coefficients
```

```
coefs <- coef(full_model_asphalt)
coefs
# Calculate intercepts based on x6 (run)
intercept_run1 <- coefs["(Intercept)"] + coefs["x61"]
intercept_runm1 <- coefs["(Intercept)"] - coefs["x61"]

# Equation when run = 1
cat("Fitted equation when run = 1:\n")
cat(paste0("y = ",
      round(intercept_run1, 4), " + ",
      round(coefs["x1"], 4), "*x1 + ",
      round(coefs["x2"], 4), "*x2 + ",
      round(coefs["x3"], 4), "*x3 + ",
      round(coefs["x4"], 4), "*x4 + ",
      round(coefs["x5"], 4), "*x5\n\n"))

# Equation when run = -1
cat("Fitted equation when run = -1:\n")
cat(paste0("y = ",
      round(intercept_runm1, 4), " + ",
      round(coefs["x1"], 4), "*x1 + ",
      round(coefs["x2"], 4), "*x2 + ",
      round(coefs["x3"], 4), "*x3 + ",
      round(coefs["x4"], 4), "*x4 + ",
      round(coefs["x5"], 4), "*x5\n"))


# Question 1 Part C
# Perform all possible subsets regression
r <- regsubsets(y ~ x1 + x2 + x3 + x4 + x5 + x6, data = asphalt)
# Extract summary of the model selection
summary_r <- summary(r)
summary_r
# View R² values for reference
```

```r
print(summary_r$rsq)
# Identify best model based on maximum R²
best_rsq_index <- which.max(summary_r$rsq)
best_rsq_index
# Get logical vector of selected variables for that model
best_vars_logical <- summary_r$which[best_rsq_index, ]
# Extract variable names (excluding intercept)
selected_vars <- names(best_vars_logical)[best_vars_logical][-1]
# Fix for factor variable name if shown as x61
selected_vars[selected_vars == "x61"] <- "x6"
# Create formula with selected predictors
reduced_formula <- as.formula(paste("y ~", paste(selected_vars, collapse = " + ")))
# Fit the model using selected predictors
best_model <- lm(reduced_formula, data = asphalt)
# View model summary
summary(best_model)

# Diagnostic Tests
# Clear any existing graphics
graphics.off()
par(mfrow = c(2, 2))
# Plot all diagnostics
plot(best_model)

# Normality Test
# Shapiro-Wilk
shapiro.test(residuals(best_model))
# Anderson-Darling
ad.test(residuals(best_model))

# Homoscedasticity Test
ncvTest(best_model)

# Autocorrelation
```

```
graphics.off()
# ACF Plot
acf(residuals(best_model), main = "ACF of Residuals")
# Durbin-Watson Test
dwtest(best_model)

# Multicollinearity Check
# VIF
vif(best_model)

# Question 2
# Load the dataset
byssinosis <- read.csv("byssinosis.csv")

# Convert predictors to factors (all are categorical)
byssinosis$Dust   <- factor(byssinosis$Dust)   # 1 = high, 2 = medium, 3 = low
byssinosis$Race   <- factor(byssinosis$Race)   # 1 = European, 2 = Other
byssinosis$Sex    <- factor(byssinosis$Sex)    # 1 = Male, 2 = Female
byssinosis$Smoke  <- factor(byssinosis$Smoke)  # 1 = Smoker, 2 = Non-smoker
byssinosis$Employ <- factor(byssinosis$Employ) # 1, 2, 3 = <10, 10–20, >20 yrs

# View structure
str(byssinosis)

# Question 2 Part A
# Fit the logistic regression model
model_logit <- glm(cbind(BysYes, BysNo) ~ Dust + Race + Sex + Smoke + Employ,
                   data = byssinosis, family = binomial)

# Model summary
summary(model_logit)

# Exponentiated coefficients (odds ratios)
exp(coef(model_logit))
```

```
# 95% confidence intervals for odds ratios
exp(confint(model_logit))

# Question 2 Part B
# Goodness-of-fit: Chi-square test on deviance
pchisq(deviance(model_logit), df.residual(model_logit), lower.tail = FALSE)

# Compute observed proportions
byssinosis$obs_prop <- byssinosis$BysYes / (byssinosis$BysYes +
byssinosis$BysNo)

# ROC curve and AUC
roc_obj <- roc(byssinosis$obs_prop, fitted(model_logit))
plot(roc_obj, main = "ROC Curve for Logistic Model")
auc(roc_obj)

# Deviance residual plot
plot(residuals(model_logit, type = "deviance"),
  ylab = "Deviance Residuals", main = "Deviance Residual Plot")
abline(h = 0, col = "red", lty = 2)
```